

Nawaraj Dahal

Palo Alto, CA

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Education

San José State University

B.S. APPLIED AND COMPUTATIONAL MATHEMATICS

GPA: 4.0/4.0

San José, California

Aug. 2025 – Expected May 2027

De Anza College

ASSOCIATE OF SCIENCE - AS, MATHEMATICS

GPA: 3.78/4.00

Cupertino, California

Sep. 2022 – Jun. 2025

Research Interests

Nonlinear Partial Differential Equations, Numerical Analysis, Scientific Computing, Mathematical Modeling

Research Experience

A Matrix-Based Perspective on Multi-Level Richardson Extrapolation

UNDERGRADUATE RESEARCHER | ADVISOR: DR. MOHAMMAD SALEEM

- Investigated multi-level Richardson extrapolation methods for numerical integration and error reduction.
- Analyzed truncation error, finite-precision effects, conditioning, and numerical stability of extrapolation schemes.
- Developed computational experiments and theoretical error analysis in MATLAB.
- Manuscript submitted to SIAM Undergraduate Research Online (SIURO).

San José State University

Sep. 2025 – Dec. 2025

Numerical Investigation of the Poisson–Boltzmann Equation

HONORS RESEARCHER | ADVISOR: DR. DANIEL BRINKMAN

- Developed and analyzed finite-difference discretizations for nonlinear Poisson–Boltzmann equations arising in semiconductor modeling.
- Implemented and compared fixed-point, Newton, and quasi-linearization methods in MATLAB.
- Investigated convergence, stability, conditioning, and iterative behavior of nonlinear numerical schemes.
- Completed an honors thesis under the supervision of Dr. Daniel Brinkman.
- Presented research at San José State University Research Day and the Department of Mathematics and Statistics End of the Year Research Showcase.

San José State University

Jan. 2026 – May 2026

Modeling and Numerical Simulation of Solar Cells

INDEPENDENT RESEARCH PROJECT

- Extending previous semiconductor drift–diffusion–Poisson modeling work to photovoltaic and solar-cell applications.
- Investigating charge transport, recombination mechanism, carrier dynamics, and photogeneration effects under illumination.
- Analyzing numerical methods for coupled nonlinear PDE systems arising in semiconductor device physics.
- Examining the influence of physical and numerical parameters on current–voltage behavior, power generation, fill factor, and device efficiency.

San José State University

May 2026 – Present

Teaching Experience

MATH 19: Precalculus

INSTRUCTIONAL STUDENT ASSISTANT: WORKSHOP FACILITATOR

- Lead workshop sessions for undergraduate students in precalculus.
- Guided students through problem-solving strategies and foundational mathematical concepts.

San José State University

Fall 2025

MATH 33LA: Differential Equations and Linear Algebra

INSTRUCTIONAL STUDENT ASSISTANT: GRADER

- Evaluated midterms and final examinations and provided written feedback to students.
- Assisted with grading logistics, rubric implementation, and assessment activities for a large-enrollment course.

San José State University

Spring 2026

Publications & Scholarly Writing

A Matrix-Based Perspective on Multi-Level Richardson Extrapolation

Submitted to *SIAM Undergraduate Research Online (SIURO)*.

2026

Honors Thesis, Department of Mathematics and Statistics, San José State University.

Posters & Presentations

San José State University Research Day

San José, California

PRESENTER FOR <NUMERICAL INVESTIGATION OF THE POISSON–BOLTZMANN EQUATION>

May 2026

- Presented research on finite-difference discretizations and iterative methods for nonlinear Poisson–Boltzmann equations arising in semiconductor modeling.
- Discussed convergence behavior, numerical implementation, and applications to semiconductor device simulation.

Department of Mathematics and Statistics End of the year celebration

San José State University

PRESENTER FOR <NUMERICAL INVESTIGATION OF THE POISSON–BOLTZMANN EQUATION>

May 2026

- Presented honors research to faculty and students during the department research showcase.
- Discussed mathematical modeling, numerical methods, and semiconductor applications.

Honors & Awards

2022–2025 Dean’s List, De Anza College

Cupertino,

California

Fall 2025 President’s List, San José State University

San José, California

Relevant Coursework

Analysis	Real Analysis, Advanced Calculus, Complex Analysis
Differential Equations	Ordinary Differential Equations and Dynamical Systems, Partial Differential Equations
Numerical Methods	Numerical Analysis I, Numerical Analysis II
Probability	Applied Probability and Statistics I
Modeling & Research	Mathematical Modeling for Life Sciences
Advanced Topics	Graduate Dynamical Systems, Graduate Partial Differential Equations(Planned), Numerical Solution of Partial Differential Equations(Planned)

Technical Skills

Scientific Computing	MATLAB
Scientific Writing	LaTeX
Programming	C++ (basic)